KS3 DIGITAL TECHNOLOGY

MACHINE LEARNING LAB

Train your own AI: collect data, build an image classifier, test how accurate it is, find out why models fail, make your data fair, and design a machine learning solution to a real problem.

LESSONS

1 Learning from examples
2 Build an image classifier
3 Is it any good?
4 When models go wrong
5 Fair data, fair AI
6 Design an ML project

LESSON 1 OF 6

LEARNING FROM EXAMPLES

In supervised learning a computer is given lots of examples with the right answers (labels).

STARTER · 5 MINUTES

DO NOW

How did you learn to tell the difference between a cat and a dog when you were little? Did someone give you a rule?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how supervised machine learning uses labelled examples, features and labels

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what supervised learning is
- ✓ I can identify features and labels in a dataset
- ✓ I can describe the training process

RULES OR EXAMPLES?

- Traditional programs: a programmer writes every rule
- Machine learning: the computer learns the rules from examples
- Hard to write rules for "is this a cat?", easy to show 10,000 cat photos
- Supervised learning: every training example comes with the right answer

KEY WORDS

MACHINE LEARNING	A type of AI where computers learn patterns from data instead of following rules written by a programmer.
SUPERVISED LEARNING	Machine learning that learns from examples labelled with the correct answers.
FEATURE	A piece of information about each example in a dataset, such as weight, colour or pixel values.
LABEL	The correct answer or category for an example, which the model learns to predict.
DATASET	A collection of examples, usually organised as a table or a set of files.

FEATURES AND LABELS

- A dataset is a collection of examples
- Features: the information about each example (size, colour, shape, pixels)
- Label: the right answer we want the model to predict ("cat", "dog")
- The model learns how features link to labels

ACTIVITY · 15 MINUTES · PAIRS

BE THE MODEL

1. Your teacher shows 12 made-up creatures labelled "Blip" or "Zorg".
2. Without being told any rules, study the examples and work out the pattern.
3. Label six new creatures and give a confidence (how sure you are, in %).
4. Write the "rule" you learned.

In your workbook, answer:

? What rule did you learn, and which features did you use?

? How is what you did like supervised machine learning?

Your workbook: Lesson 1 · tick each step off as you go

TRAINING AND PREDICTING

- Training: the model adjusts itself to match features to labels
- It keeps adjusting until it makes few mistakes on the examples
- Predicting: it labels a new example it hasn't seen
- It gives a confidence score, e.g. "cat: 92%"

QUICK CHECK

1. What is supervised learning?

A Learning from examples that have the right answers (labels)
B Learning without any data
C Learning with a teacher watching the computer
D Writing every rule by hand

2. In a dataset of fruit, "weight" is a...

A prediction
B label
C model
D feature

3. In a dataset of fruit, "apple" or "orange" is the...

A feature
B label
C algorithm
D dataset

4. What does a confidence score of 92% mean?

A The model is right 92% of the time on everything
B 92 people agreed
C The model is fairly sure of its prediction
D The image is 92% clear

QUICK CHECK: ANSWERS

1. **A: Learning from examples that have the right answers (labels)**
   Labelled examples supervise the learning.

2. **D: feature**
   Features describe each example.

3. **B: label**
   The label is the answer to predict.

4. **C: The model is fairly sure of its prediction**
   It's how confident the model is about that prediction.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain the difference between a feature and a label in a dataset.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what supervised learning is
- ✓ I can identify features and labels in a dataset
- ✓ I can describe the training process

LESSON 2 OF 6

BUILD AN IMAGE CLASSIFIER

A classifier sorts inputs into categories.

MACHINE LEARNING LAB

STARTER · 5 MINUTES

DO NOW

You want an AI to tell the difference between a thumbs up and a thumbs down. What photos would you need to give it?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Collect training data and train an image classifier

SUCCESS CRITERIA: I CAN...

- ✓ I can collect varied training data for each class
- ✓ I can train an image classifier
- ✓ I can explain how the data affects the model

WHAT'S A CLASSIFIER?

- A classifier is a model that sorts things into categories, called classes
- Spam or not spam; cat, dog or rabbit; recycling or general waste
- An image classifier does this with pictures
- Teachable Machine lets you build one in your browser in minutes

KEY WORDS

CLASSIFIER	A model that sorts inputs into categories.
CLASS	One of the categories a classifier can choose between.
TRAINING DATA	The labelled examples a model learns from.

GOOD DATA BEATS CLEVER CODE

- At least 30-50 examples per class to start
- Vary it: angles, distances, lighting, backgrounds, different hands or objects
- Keep classes balanced: a similar number of examples in each
- Add a "nothing" class so it doesn't have to guess when nothing is shown

TRAIN, TRY, IMPROVE

- Train: the tool learns patterns from your examples
- Try it live with new examples
- Where does it go wrong? Add more varied data there
- Retrain and compare

ACTIVITY · 30 MINUTES · PAIRS

RECYCLING SORTER

1. Open Teachable Machine and create an Image Project with three classes: Paper, Plastic and Nothing.

2. Collect at least 30 varied images for each class using classroom objects (cards, plastic bottles, an empty desk).

3. Train the model and test it with objects you did NOT use in training.

4. Find one situation where it fails, add data to fix it and retrain.

In your workbook, answer:

? How did you make your training data varied?

? Describe one mistake your model made, why you think it happened and how you fixed it.

Your workbook: Lesson 2 · tick each step off as you go

QUICK CHECK

1. What does a classifier do?

A Speeds up the internet
B Writes stories
C Sorts inputs into categories
D Cleans the computer

2. Which training data is best for a hand-gesture classifier?

A Ten identical photos
B Photos of only one person's hand in one place
C Many photos with different hands, angles and lighting
D One photo per class

3. Why add a "nothing" class?

A So the model doesn't have to guess a class when nothing is shown
B To delete wrong images
C Because the tool needs four classes
D To make training slower

4. Your model fails in dim light. What should you do?

A Change its colour
B Delete the model
C Only use it at night
D Add more training examples in dim light and retrain

QUICK CHECK: ANSWERS

1. **C: Sorts inputs into categories**
Classifiers assign classes.

2. **C: Many photos with different hands, angles and lighting**
Variety helps the model generalise.

3. **A: So the model doesn't have to guess a class when nothing is shown**
It gives the model a valid "none" answer.

4. **D: Add more training examples in dim light and retrain**
Fill the gap in the data.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three things that make training data good for an image classifier.

Look back at today's success criteria: can you tick them all?

- ✓ I can collect varied training data for each class
- ✓ I can train an image classifier
- ✓ I can explain how the data affects the model

LESSON 3 OF 6

IS IT ANY GOOD?

A model must be tested on data it has never seen.

DO NOW

If a teacher gave you the exact exam questions to practise, then the same questions in the exam, would your score show what you really know?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Test a model using separate test data, accuracy and a confusion matrix

SUCCESS CRITERIA: I CAN...

- ✓ I can explain why test data must be separate from training data
- ✓ I can calculate accuracy
- ✓ I can read a confusion matrix

DON'T MARK YOUR OWN HOMEWORK

- Testing on training data is like sitting an exam you've seen the answers to
- Split the data: usually about 80% for training, 20% for testing
- Test data must never be used in training
- It tells you how the model will do on brand-new examples

KEY WORDS

TEST DATA	Examples kept separate from training and used to check how well a model works on new data.
ACCURACY	The percentage of a model's predictions that are correct.
CONFUSION MATRIX	A table comparing actual and predicted classes, showing which classes a model mixes up.

ACCURACY

- Accuracy = correct predictions ÷ total predictions × 100
- 45 correct out of 50 = 90% accuracy
- But accuracy alone can hide problems
- 95% accurate at spotting a rare disease might still miss most ill people

CONFUSION MATRIX

- A table comparing actual classes with predicted classes
- The diagonal shows correct predictions
- Everything else shows which classes get confused
- Helps you decide what data to add next

ACTIVITY · 25 MINUTES · PAIRS

TEST YOUR SORTER

1. Gather 10 NEW test objects for each class from your Recycling sorter (lesson 2) that were not used in training.
2. Test each one and record the actual class and the predicted class.
3. Fill in a confusion matrix and calculate the accuracy.
4. Decide which classes get confused and what data you would add.

In your workbook, answer:

? What was your model's accuracy? Show your working.

? Which two classes were most often confused, and what would you do about it?

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. Why must test data be kept separate from training data?

A Because the tool deletes it
B To see how the model does on examples it hasn't seen
C To make training faster
D It doesn't need to be

2. A model gets 18 out of 20 test items right. What is its accuracy?

A 95%
B 90%
C 18%
D 80%

3. In a confusion matrix, where are the correct predictions?

A In the top row
B On the diagonal
C In the bottom corner
D In the last column

4. Why can accuracy be misleading?

A A model can score high by always guessing the most common class
B It only works for images
C Accuracy is always wrong
D It is measured in kilograms

QUICK CHECK: ANSWERS

1. B: To see how the model does on examples it hasn't seen

Unseen data shows real performance.

2. B: 90%

18 ÷ 20 × 100 = 90%.

3. B: On the diagonal

Correct = actual matches predicted.

4. A: A model can score high by always guessing the most common class

Rare classes can hide failure.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

A model is tested on 50 images and gets 42 right. Calculate its accuracy.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain why test data must be separate from training data
- ✓ I can calculate accuracy
- ✓ I can read a confusion matrix

LESSON 4 OF 6

WHEN MODELS GO WRONG

Models fail when their data doesn't match the real world: too few examples, not enough variety, unusual edge cases, or overfitting, where a model memorises its training data instead of learning general patterns.

STARTER · 5 MINUTES

DO NOW

A model trained to spot wolves got 100% on its test. But it was actually spotting snow. How could that happen?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain common reasons machine learning models fail, including overfitting, too little data and edge cases

SUCCESS CRITERIA: I CAN...

- ✓ I can explain overfitting
- ✓ I can explain why models struggle with edge cases
- ✓ I can suggest how to make a model more robust

THE WOLF AND THE SNOW

- A famous research example: a model learned to tell wolves from huskies
- Most wolf photos had snow in the background
- The model learned "snow = wolf", not what a wolf looks like
- This is called shortcut learning

KEY WORDS

OVERFITTING	When a model learns its training data too closely and performs badly on new data.
EDGE CASE	An unusual situation that rarely appears in the training data.
GENERALISE	To work well on new examples, not just the ones used in training.
SHORTCUT LEARNING	When a model uses an accidental clue (like a background) instead of the real feature.

OVERFITTING

- Overfitting: the model memorises the training examples, including their quirks
- It scores brilliantly on training data but badly on new data
- Like learning answers by heart without understanding
- Fixes: more varied data, simpler models, proper testing

EDGE CASES

- Edge cases: unusual situations rarely seen in training
- A self-driving car's AI might not recognise a person in a dinosaur costume
- Real life is full of edge cases
- That's why safety-critical AI needs huge testing and human oversight

ACTIVITY · 25 MINUTES · PAIRS

BREAK YOUR MODEL

1. Take your trained Recycling sorter from lesson 2.
2. Try to make it fail: change the background, the lighting, the distance or use unusual objects.
3. Record each attempt and whether it fooled the model.
4. Explain each failure: shortcut learning, overfitting, too little data or an edge case?

In your workbook, answer:

? Describe your best "fail" and explain why it happened.

? Suggest two changes to make your model more robust.

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. What is overfitting?

A When a model is too small
B When the computer overheats
C When the data is too varied
D When a model memorises its training data and does badly on new data

2. Training accuracy 100%, test accuracy 60%. What is the likely problem?

A Too much test data
B Perfect model
C Overfitting
D Underfitting

3. What is an edge case?

A The border of an image
B The first example in a dataset
C A type of computer case
D An unusual situation rarely seen in training

4. The wolf model actually learned to spot snow. This is…

A overfitting to colours
B shortcut learning
C an edge case
D accuracy

QUICK CHECK: ANSWERS

1. **D: When a model memorises its training data and does badly on new data**
   Overfit models don't generalise.

2. **C: Overfitting**
   A big gap between training and test suggests overfitting.

3. **D: An unusual situation rarely seen in training**
   Edge cases are rare, unusual inputs.

4. **B: shortcut learning**
   It used an accidental clue.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain what is meant by overfitting.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain overfitting
- ✓ I can explain why models struggle with edge cases
- ✓ I can suggest how to make a model more robust

LESSON 5 OF 6

FAIR DATA, FAIR AI

If training data under-represents some groups or repeats past unfairness, the model will be less accurate or unfair for those groups.

MACHINE LEARNING LAB

DO NOW

A soap dispenser works for some people's hands but not others. What might have gone wrong when it was designed and tested?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how biased data leads to unfair machine learning and how to reduce bias

SUCCESS CRITERIA: I CAN...

- ✓ I can explain how bias gets into data
- ✓ I can describe real examples of unfair AI
- ✓ I can suggest ways to reduce bias

UNFAIR IN, UNFAIR OUT

- Models learn from data, so they learn the data's gaps and unfairness
- Under-representation: some groups barely appear in the data
- Historical bias: data from the past reflects past unfairness
- Labelling bias: the people labelling data add their own assumptions

KEY WORDS

BIAS	When a system is less accurate or less fair for some people, often because of its data.
REPRESENTATIVE DATA	Data that includes a fair mix of all the people and situations a system will meet.
FAIRNESS	Making sure a system does not treat people worse because of who they are.

REAL EXAMPLES

- Research found some face-analysis systems were far less accurate for darker-skinned women than for lighter-skinned men
- A company stopped using a CV-sorting AI that learned to prefer men, because past hires were mostly men
- Speech recognition has often worked less well for some accents
- Health tools trained mostly on one group may miss illness in others

MAKING AI FAIRER

- Check who is in the data: is it representative of everyone who will use it?
- Test accuracy separately for different groups
- Involve diverse people in design and testing
- Be open about limits; keep humans in charge of big decisions

ACTIVITY · 20 MINUTES · GROUPS OF 3

DATASET AUDIT

1. Your teacher gives you a summary of a (fictional) dataset for a school "smart door" that recognises pupils' faces.

2. Find who is under-represented in the data.

3. Predict which pupils the system might fail for and what problems this could cause.

4. Write three recommendations to make it fairer, or argue that it shouldn't be used at all.

In your workbook, answer:

? Which groups were under-represented, and what could go wrong for them?

? Write your three recommendations (or your argument against the system).

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. Where does bias in machine learning usually come from?

A The speed of the processor
B The programmer's handwriting
C The training data
D The colour of the computer

2. What is under-representation?

A When some groups barely appear in the data
B When data is encrypted
C When labels are correct
D When data is too big

3. Why did a CV-sorting AI learn to prefer men?

A It was programmed to
B It learned from past hiring data that was mostly men
C The computer was broken
D Women didn't apply

4. Which step helps reduce bias?

A Using less data
B Hiding how the AI works
C Only testing overall accuracy
D Testing accuracy separately for different groups

QUICK CHECK: ANSWERS

1. **C: The training data**
   Models learn data's unfairness.

2. **A: When some groups barely appear in the data**
   Missing groups lead to poor accuracy for them.

3. **B: It learned from past hiring data that was mostly men**
   Historical bias came from the data.

4. **D: Testing accuracy separately for different groups**
   Group-level testing reveals unfairness.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how a face-recognition system could end up less accurate for some groups of people.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain how bias gets into data
- ✓ I can describe real examples of unfair AI
- ✓ I can suggest ways to reduce bias

LESSON 6 OF 6

DESIGN AN ML PROJECT

Real ML projects start with a clear problem, then plan the data, how to train and test, how success will be measured, and the risks.

MACHINE LEARNING LAB

STARTER · 5 MINUTES

DO NOW

Think of one annoying problem at school. Could an AI help? What would it need to learn from?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Plan a machine learning solution to a real problem, including data, testing and ethics

SUCCESS CRITERIA: I CAN...

- ✓ I can describe the stages of an ML project
- ✓ I can plan the data and testing for a model
- ✓ I can identify the ethical risks of my solution

THE ML PROJECT CYCLE

- 1 Define the problem: what exactly should the AI predict or classify?
- 2 Collect and label data, with permission
- 3 Train, then test on separate data
- 4 Evaluate, improve, and keep monitoring after launch

KEY WORDS

PROBLEM STATEMENT	A clear description of the problem a project will solve, and for whom.
EVALUATION	Judging how well a solution works against success criteria.
HUMAN OVERSIGHT	People checking an AI's decisions and being able to overrule them.

ASK THE HARD QUESTIONS

- Is AI actually needed, or would a simple rule work?
- Where will the data come from, and do we have permission?
- What happens when it's wrong, and who is affected?
- Who checks its decisions (human oversight)?

SUCCESS CRITERIA

- Decide in advance: what accuracy is good enough?
- Which mistakes are worse? (missing a problem vs a false alarm)
- Test with real users in real conditions
- Plan to retrain as things change

ACTIVITY · 35 MINUTES · GROUPS OF 3

PITCH AN AI FOR OUR SCHOOL

1. Choose a school problem (e.g. sorting lost property, identifying birds for the eco-club, spotting litter in photos of the playground).

2. Complete a project plan: problem statement, classes, what data you'd collect and how, how you'd test it, your success criteria.

3. Complete the ethics checklist for your idea.

4. Pitch it to the class in 2 minutes. The class votes: build it, change it, or don't use AI.

In your workbook, answer:

? Write your problem statement and the classes your model would predict.

? Describe your data, testing plan and success criteria.

Your workbook: Lesson 6 · tick each step off as you go

? What is the biggest risk of your idea, and how would you reduce it?

QUICK CHECK

1. What is the first stage of an ML project?

A Deploy the app
B Train the model
C Define the problem clearly
D Buy a faster computer

2. When might a normal program be better than ML?

A When a simple rule solves the problem
B Never
C When recognising images
D When patterns are too complex for rules

3. What is human oversight?

A People checking and able to overrule the AI's decisions
B A camera watching humans
C A faster algorithm
D A type of training data

4. Why decide success criteria before testing?

A It doesn't matter when
B So testing is faster
C Because the AI needs them
D So you can judge the results fairly

QUICK CHECK: ANSWERS

1. C: Define the problem clearly

Start with a clear problem statement.

2. A: When a simple rule solves the problem

Simple rules are cheaper and clearer.

3. A: People checking and able to overrule the AI's decisions

Humans stay in control.

4. D: So you can judge the results fairly

Deciding first keeps evaluation honest.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

A school wants AI to mark pupils' handwritten spelling tests. Identify three questions the school should ask before building it.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe the stages of an ML project
- ✓ I can plan the data and testing for a model
- ✓ I can identify the ethical risks of my solution

END OF TOPIC

MACHINE LEARNING LAB

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.